



ISO 7218:2024 in official control food microbiology laboratories: key revisions, quality implications, operational challenges, implementation considerations, and opportunities for harmonization

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Abstract

The updated version of ISO 7218:2024, titled “Microbiology of the food chain—General requirements and guidance for microbiological examinations”, serves as a significant horizontal standard that underpins microbiological testing throughout the food chain. The 2024 update not only involves editorial refinements but also incorporates a clearer risk-based approach to sample management, contamination prevention, internal quality assurance, quantitative result handling, and laboratory reporting. These modifications are particularly important for official control microbiology laboratories, where analytical outcomes must be scientifically sound, operationally traceable, and sufficiently reliable to support regulatory decision-making. Official control food microbiology laboratory refers to any public or private laboratory formally designated by the competent authority, pursuant to Article 37 of Regulation (EU) 2017/625, to perform microbiological analyses on samples collected during official controls or other official activities. This manuscript reviews the scope and role of ISO 7218:2024 and critically discusses the principal revisions introduced by the new edition. Particular attention is given to harmonized transport and storage conditions, the provision of freezing before examination, the limitations of retesting stored samples, reinforced laboratory zoning and hygiene requirements, expanded process-oriented internal quality control, quantitative counting rules, broader acceptance of alternative confirmation and identification technologies, and reduced testing for ready-to-use media under controlled conditions. This revision is considered to reflect a shift toward a system-centric quality assurance approach rather than a methodology-centric approach to microbiology. The successful implementation of this development will depend on whether various parameters are met, such as the availability of personnel, equipment and infrastructure, work planning, the adequacy and supervision of suppliers, and a comprehensive risk assessment. Therefore, ISO 7218:2024 can enhance credibility, transparency, and standardization of food microbiology and should be regarded as more than merely a technical update.

Keywords Food microbiology laboratories · Food safety · ISO 7218:2024 · ISO/IEC 17025 · Internal quality control · Cross-contamination · Sample handling

Introduction

In microbiological testing, laboratory quality depends on the ability to generate results that are accurate, reliable, and available in time to support appropriate decisions. Reliability is not determined only by the analytical method, but by the controlled performance of the entire examination process, beginning with representative sampling and continuing through sample transport, storage, preparation, contamination prevention, equipment performance, internal quality control (IQC), and reporting. Sampling errors and inadequate consideration of matrix heterogeneity may substantially affect the uncertainty and interpretation of the

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final result, even where sampling is performed outside the laboratory's direct responsibility.

The revised standard also accommodates greater automation of defined-volume handling, including automated dilution and pipetting systems and spiral plating devices, provided that their performance is appropriately verified, monitored, and documented. Since every measurement process includes some degree of uncertainty, the role of a quality system is to minimize errors and control the factors that may compromise result validity. Even a seemingly small error rate can have important consequences in high-volume laboratory settings, particularly in official control laboratories where results may support regulatory or legal action [1]. A Laboratory Quality Management System (LQMS) can be understood as a structured framework through which laboratory management defines, documents, and monitors the personnel, procedures, and processes required to ensure consistent quality in laboratory activities, generated outputs, and reported results. In this context, laboratory quality is shaped by the entire workflow, which begins with customer requirements and extends through testing, reporting, interpretation, and the decisions or actions that follow. This workflow includes not only the analytical procedure itself, but also all pre-testing and post-testing activities that may influence the reliability and usefulness of laboratory results. According to WHO [1], the laboratory workflow is commonly divided

into three interconnected phases: the pre-analytic, analytic, and post-analytic phase. Figure 1 illustrates this system-wide perspective, which is relevant to the ISO 7218:2024 standard and which enhances coordinated control of the entire examination process [2, 3].

Official food laboratories play a crucial role in protecting public health and ensuring the effective operation of food control systems. Regular surveillance, regulatory enforcement, epidemic investigation, and the evaluation of conformity with microbiological standards are all supported by their analytical findings. Laboratory results must consequently be technically sound, repeatable, traceable, and robust enough to support administrative or legal decision-making in official control situations. The integrity of the entire examination process, including sample transport, receipt, storage, preparation, incubation, calculation, interpretation, and reporting, is just as important to the validity of microbiological testing as the analytical method itself [2–4].

Official food laboratories are the foundation of evidence-based public health decision-making within the framework of official controls, as stated by Articles 34 and 35 of Regulation (EU) 2017/625 [4]. The necessity for technical excellence is strengthened by the switch to the revised ISO 7218:2024 standard, which requires that analytical data are characterized by uninterrupted traceability, reproducibility, and statistical validity [2].

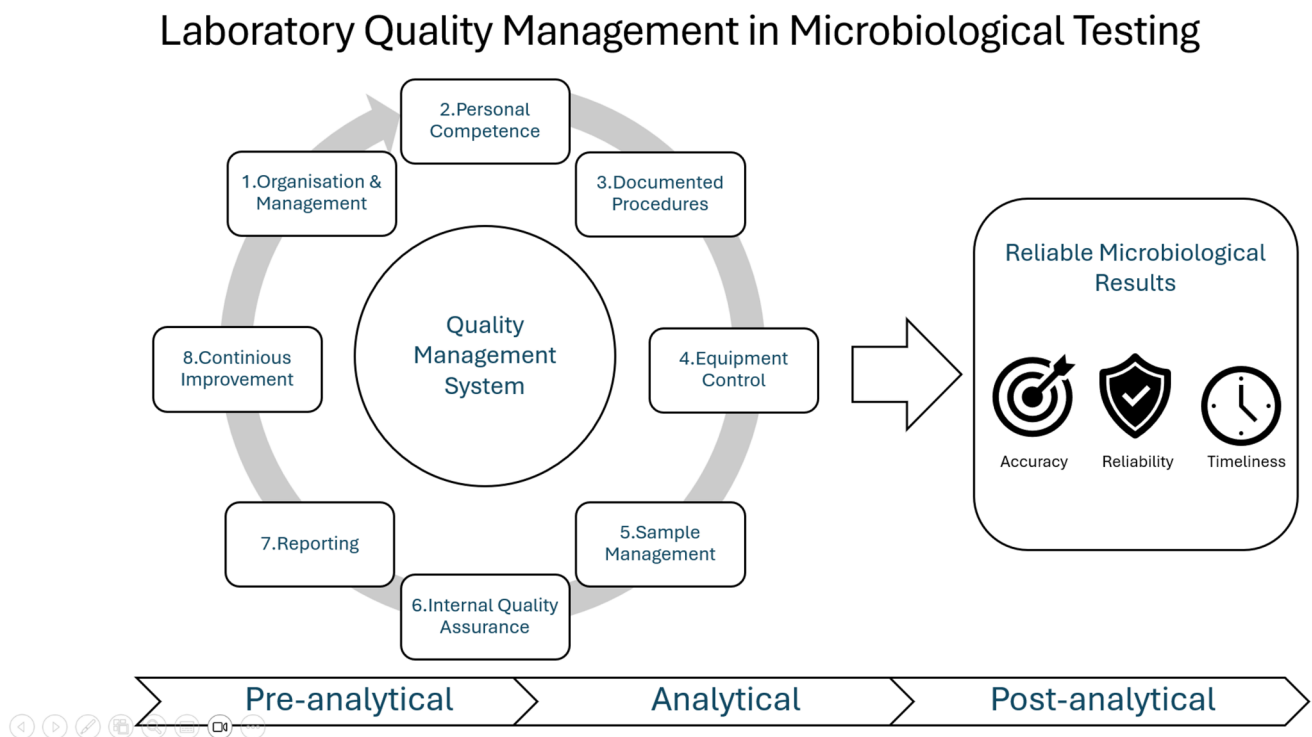


Fig. 1 Schematic representation of a laboratory quality management system in microbiological testing based on the principles of ISO 7218:2024 and ISO/IEC 17025:2017

In accordance with the requirements of the new standard, the result validity is not confined to the analytical phase but depends on the appropriate management of the entire examination process. It includes strict standardization of controlled pre-analytical stages, such as sample transport and preparation, as well as post-analytical procedures, including calculation, interpretation, measurement uncertainty where applicable, and reporting of the results, ensuring that the findings are sufficiently robust to support judicial or administrative actions. Microbiological examination presents a distinct technical challenge because laboratories must ensure that only microorganisms present in the sample are detected or enumerated and that the examination procedure does not introduce other microorganisms. For this reason, good laboratory practice in food microbiology extends beyond analytical performance to encompass staff hygiene, aseptic handling, environmental control, appropriate equipment management, biosafety awareness, and quality assurance measures applied throughout the testing process. The validity of the result is therefore inseparable from the validity of the laboratory conditions under which the result is produced [2, 3, 5].

Within this framework, ISO 7218:2024 provides the general technical basis for microbiological examinations across the food chain. ISO 7218 has long provided general guidance for food microbiology laboratories, with the 2024 edition updating the 2007 version and its 2013 amendment with a stronger process- and risk-based approach. The revised edition replaces ISO 7218:2007 and its amendment while introducing important changes in areas such as result calculation, equipment organization, internal quality control, cross-referencing with related microbiological standards, and the characterization of control microorganisms [2, 6, 7].

The purpose of the present paper is to examine the scope and role of ISO 7218:2024 in food microbiology, discuss the principal revisions introduced by the new edition, and assess how these changes may affect laboratory processes. Special attention is given to the practical implications of the new standard regarding laboratory workflow, sample handling, prevention of contamination, internal quality control, quantitative microbiology, and reporting. In this sense, ISO 7218:2024 may be understood not simply as a revised technical text, but as an operational framework for strengthening reliability, consistency, and transparency in official microbiological testing [2–4].

To the best of our knowledge, this paper is among the first bibliographic and normative assessments of the implementation of ISO 7218:2024 as applied to official food microbiological control laboratories. ISO 7218:2024 not only specifies general requirements but also provides guidance for microbiological analyses in the food chain and replaces the previous edition ISO 7218:2007, as well as its amendment in 2013 [2, 6, 7]. The particular contribution of the

article is that it highlights this not only as a model of laboratory operations but also as an operational framework for enhancing laboratory quality in official food laboratories. This specific standard is directly linked to the application of specific horizontal and vertical microbiological methods with technical compliance with the requirements of ISO/IEC 17025, as well as with good laboratory practices in laboratories that analyze samples from the food chain [2, 3]. Thus, this article contributes substantially to the understanding of the new edition of ISO 7218:2024 in the systematic management of the critical stages of microbiological examination in the critical stages of receipt, from receipt to sample preparation to analysis and finally to the interpretation and reporting of results. At the same time, the work places the application of ISO 7218:2024 in the broader context of official controls. Regulation (EU) 2017/625 is the main EU legislative framework for official controls and other official activities aimed at ensuring the application of law on food, feed, animal health and welfare, plant health, and plant protection products [4]. Therefore, the application of ISO 7218:2024 in official control laboratories is not simply a matter of technical compliance but is linked to the reliability of data used for public health decision-making, enforcement of legislation, and harmonization of controls in the agri-food chain.

Methodology

The central research questions of the paper can be formulated as follows: what are the main qualitative, operational, and regulatory implications of the implementation of ISO 7218:2024 in official food microbiology laboratories and how may these changes affect in terms of reliability, harmonization, and management. On a second level, the paper explores how the new standard can support the technical competence of food microbiology laboratories, compliance with ISO/IEC 17025, the correct application of specific microbiological methods, and the strengthening of an integrated laboratory quality management system [1, 3]. Methodologically, the paper was designed as a Perspective Review based on a critical analysis of ISO 7218:2024, ISO 7218:2007, and its 2013 amendment, legislation, and relevant literature. Specifically, an approach was followed that included the collection, study and interpretative analysis of data from international standards, regulatory texts, and relevant literature on quality in food microbiological laboratories. The central object of the analysis was ISO 7218:2024, which was examined in terms of its scope, its technical and operational requirements, as well as its relationship with the previous edition ISO 7218:2007 and its amendment [2, 6, 7]. The comparison of the two editions allowed the capture of the main changes and the

assessment of their importance for the daily practice of laboratories carrying out official microbiological controls in the food chain. The analysis also included the connection of ISO 7218:2024 with ISO/IEC 17025:2017, given that official control laboratories must operate in a manner that documents reliability, technical competence, as well as the defensible production of results. In addition, the relevance of the standard to the principles of the laboratory quality management system, as described by the World Health Organization, was examined, according to which quality is not limited to the analysis phase, but concerns the entire workflow, including the pre-analytical, analytical, and post-analytical stages [1–3]. In this light, the article highlights that ISO 7218:2024 enhances a more integrated and systemic view of laboratory quality in food microbiology. It is important to clarify that this work was not based on qualitative research with interviews, nor on quantitative research with questionnaires, but this Perspective Review was based on a critical reading, understanding and comparison of standards, relevant legislation, and literature, with the changes introduced by the new version of ISO 7218 interpreted in relation to the operational context of official control laboratories. In this way, the article acts as a bridge between the text of the standard and its practical application in laboratory environments where the reliability of microbiological results has a direct impact on public health, compliance with legislation, and the harmonization of official controls.

Scope and role of ISO 7218:2024 in food microbiology

ISO 7218:2024 is a horizontal standard that specifies general requirements and provides guidance for microbiological examinations performed across the food chain. Its scope includes (a) the implementation of specific horizontal or vertical international standards for the detection or enumeration of microorganisms, (b) good laboratory practices for the examination of samples originating from the food chain, and (c) technical guidance on compliance with ISO/IEC 17025. The standard therefore occupies a foundational position within food microbiology, functioning as a general framework that supports the correct and consistent application of more specific reference methods [2, 3].

The field of application of ISO 7218:2024 is broad. Under the standard, the methods apply to testing bacteria, yeasts, and molds and can also be used for parasites and viruses when accompanied by additional specific guidance. The scope includes the microbiology of the entire food chain, from primary production through food and feed products, as well as the facilities where these are produced and handled. It also

includes water used during food production or classified as food under national law. At the same time, the standard explicitly excludes examinations for toxins or other microbial metabolites, thereby clarifying the boundaries of its intended use [2].

A particularly important conceptual contribution of ISO 7218:2024 is its distinction between horizontal methods, vertical methods, specific standards, and general standards. A horizontal method is broadly applicable across matrices within the food chain, whereas a vertical method is specific to a product or sector. A specific standard describes the reference method for the examination of a target microorganism or group of microorganisms, while the general standard provides the general guidance and requirements necessary for the correct implementation of those specific standards. This distinction has practical value, as it explains why ISO 7218:2024 should not be applied exclusively as an analytical method, but as a technical and organizational basis that can ensure consistency across a wide range of microbiological tests. [2].

Furthermore, the role of this new ISO standard is further strengthened by the fact that it incorporates other ISO standards relevant to food microbiology. The standard explicitly refers to ISO 22174 for PCR-based tests and is also linked to other standards covering culture media performance, measurement uncertainty, proficiency testing by interlaboratory comparison, colony-counting procedures, and the validation of alternative methods. ISO 7218:2024, by specifying general requirements applicable to all microbiological methods, helps to align the analytical standards of specific methods with the requirements of accreditation and quality assurance of official microbiological control laboratories. For example, it links the requirements of ISO 11133 on the performance of culture media, ISO 19036 on measurement uncertainty, ISO/TS 22117 on proficiency testing, and the ISO 16140 series on method validation with the laboratory's quality management practice [2, 3, 8–16]. For official food microbiology laboratories, the role of ISO 7218:2024 is extremely important. These laboratories must produce results that are comparable over time and between institutions, while operating under conditions that can withstand accreditation inspection, accreditation review, and regulatory control. ISO 7218:2024 supports the harmonization of laboratory practice and improves the reproducibility and defensible nature of microbiological results by defining general requirements for facilities, personnel, hygiene, sample handling, equipment, process control, and reporting of results. At the same time, the standard repeatedly emphasizes risk analysis and laboratory adaptation, recognizing that harmonization must be achieved through consistent application of the standard rather than through absolute uniformity of procedures [2–4].

Key revisions introduced by ISO 7218:2024 and their implications for official control microbiology laboratories

From descriptive guidance to risk-oriented laboratory governance

Although all versions of ISO 7218 have traditionally been a general technical standard for microbiological testing up until 2024, the 2024 revision appears to extend beyond the harmonization of analytical procedures. The 2024 version is a clear model for risk-centered laboratory management [2, 17]. The changes highlighted in the foreword—simplified calculations, added calculators, reorganization of equipment requirements, stronger cross-referencing to related standards, and expanded treatment of laboratory quality control—should not be seen as merely editorial refinements. Taken together, these revisions reinforce and make more explicit the process-oriented and risk-based control of factors affecting result validity, an objective that was already central to ISO 7218:2007 [2].

This shift is particularly important for official control laboratories. In such settings, the analytical result is not only a scientific output but also a regulatory instrument. Consequently, the real value of ISO 7218:2024 lies less in the novelty of isolated technical clauses and more in its attempt to formalize those parts of laboratory practice that historically depended on local experience, unwritten conventions or variable professional judgment. Laboratories are no longer expected merely to apply good practice; they are expected to show that the risks affecting result validity have been identified, documented, monitored, and, where necessary, minimized [2–4]. In the author's assessment the simplified calculations, the ability to use Excel applications to verify correct calculations (e.g., the use of the calculators available at <https://standards.iso.org/iso/7218/ed-4/en/>) and the extensive treatment of laboratory quality control are not considered simply technical improvements compared to the previous version of the standard [2–4].

Changes in transport and storage conditions between ISO 7218:2007 and ISO 7218:2024

The harmonization of temperature requirements for the transport of samples and their storage in the laboratory before analysis is one of the clearest points of the stricter enhanced control provided for by the 2024 standard (Table 1). Clear temperature categories are provided for shelf-stable products, products not stable at room temperature, frozen products, and environmental surface samples, thereby reducing ambiguity in interpretation and

Table 1 Harmonization of temperatures in transportation and storage conditions before analysis

Product category	Transport		Storage before analysis	
	ISO 7218:2007	ISO 7218:2024	ISO 7218:2007	ISO 7218:2024
Shelf-stable products	Ambient temperature (< 40 °C)	Room temperature (18 to 27 °C)	Room temperature (18 to 27 °C)	
Frozen or deep-frozen products	Below – 15 °C, preferably below – 18 °C			
Other products not stable at room temperature	1 °C—8 °C	5 °C ± 3 °C	3 °C ± 2 °C	5 °C ± 3 °C
Surface samples (swab samples)	See ISO 18593 and ISO 17604	Between 1 °C and 8 °C; see ISO 18593 and ISO 17604	See ISO 18593 and ISO 17604	Between 1 °C and 8 °C; see ISO 18593 and ISO 17604

The transport conditions apply during the transport of the samples to the laboratory (clause 9.2), while clause 9.4.2 refers to storage prior to examination and specifies the recommended short-term storage temperatures. The temperatures stated are to be observed unless otherwise required by the relevant specific standard

promoting more consistent sample handling during transport and before analysis. The revised standard provides clearer and more harmonized temperature categories for the transport and storage of different sample types [2, 10–12]. Although the general distinction between stable and unstable products was already present in ISO 7218:2007, the 2024 edition reduces ambiguity and supports more consistent application and documentation of transport conditions. The 2024 edition introduces some clarifications to transport and storage conditions and reinforces the need to limit the time elapsed between sampling and microbiological examination, thereby supporting more consistent pre-analytical handling. Logistical investment is not inherently required; however, additional equipment or revised transport arrangements may be necessary where existing systems cannot demonstrate compliance with the specified transport conditions (Table 1) [2, 4].

Freezing before examination: scientifically justified but operationally disruptive

One of the important changes in the new standard is that freezing samples except for the already frozen products before testing is not permitted since it can affect test results by changing the microbial load [2]. This is scientifically justified given that the load changes due to the fact that microbiological analyses are particularly sensitive to temperature fluctuations, which probably affects the viability and distribution of the population. This provision, as stated in clause 9.3, clearly refers to the possible effect of freezing on the microbial load and, in this way, supports the validity of the test results.

However, this provision's operational consequences are what make it significant. In the authors' assessment, this clarification has important operational implications for capacity planning and design. In reality, laboratories can no longer rely on storage as a regular compensatory tool for examination delays due to the absence of freezing's ability to be employed as a fallback alternative. This can include reconsidering and reevaluating receiving days, scheduling analyses, giving particularly perishable matrices priority, and organizing laboratory workflow. The clause is a test of whether the laboratory's working model is actually in line with microbiological validity for official control laboratories, rather than just a technical modification. According to the authors' assessment, capacity planning and proactive control of pre-analytical conditions are required rather than post-analytical correction, for the application of this constraint [2, 4].

Retesting of stored samples: strengthening scientific defensibility over administrative convenience

According to this standard, retesting of stored samples is not an acceptable practice, given that the microbial distribution in the samples is heterogeneous and the microbial population may change during storage, potentially resulting in different quantitative and qualitative results. Similarly, the 2007 edition advised against, but did not explicitly prohibit, the retesting of samples [7]. However, this possibility of re-examination can be given strictly in special cases, such as the need for epidemiological investigation. When a client seeks confirmation of a non-compliant result, requests for re-examination may arise. However, microbiological samples generally cannot be treated in the same way as stable, well-homogenized samples used in some chemical analyses.

A second result obtained from a stored sample is therefore not simply a confirmation or invalidation of the first but may reflect changes in microbial viability or the heterogeneous distribution of microorganisms within the product. If a stored sample is re-analyzed, the standard requires that it be separately reported and recorded as a re-examination of the sample, thereby enhancing the interpretative transparency of the result. In ideal circumstances, a new sample of the same product may be requested. This phrase safeguards microbiology's scientific rationale from being construed as chemical re-analysis of a homogeneous sample. A second result produced from a preserved sample is not just a confirmation or contradiction of the first; it could indicate changing microbial dispersion, viability change, or sampling heterogeneity [18]. The standard requires retest findings to be clearly defined and reported separately, improving interpretive clarity and legal defensibility. It may also, however, create tension with regulatory expectations for confirmatory repetition, thereby requiring stronger scientific communication between laboratories and competent authorities [2, 4].

Workflow segregation and contamination prevention: from good practice to auditable expectation

ISO 7218:2024 provides clear guidance on laboratory layout, contamination prevention, and the spatial or temporal separation of laboratory activities. There is detailed guidance for the analyst regarding the separation of activities. There are points such as the receipt of samples, their preparation, the dilution process, their coating on nutrient media, how pathogens will be handled, the tasks related to the preparation of nutrient media, the sterilization of nutrient media, and in addition the spatial and temporal separation based on the available infrastructure [2, 5, 9]. In official laboratories following the standard, it is specified, for example, that

the contact time between sample preparation and contact with the substrate should ideally be within 20 min and certainly not exceed a maximum of 45 min. This means that special care should be taken during inspections, that there is appropriate documentation for the time that has elapsed, and obviously that smaller sample series should be used for the procedure in question, in order to achieve the time limits. In well-equipped laboratories, these requirements are implemented to the extent that there may be a structural gap for smaller laboratories where there should be adequate documentation for the application of these criteria [2, 5].

Expanded internal quality control provisions

The treatment of IQC is undoubtedly one of the significant changes in the ISO 7218:2024 standard. Although ISO 7218:2007 already included quality control requirements, ISO 7218:2024 presents IQC in a more explicit, structured, process-oriented, and risk-based manner, with the IQC program reflecting the frequency of each test performed and being proportionate to its potential impact on the validity of laboratory results. The main differences between the quality control provisions of ISO 7218:2007 and the more detailed provisions of ISO 7218:2024 are summarized in Table 2. This standard now links microbiological tests and the validity of the procedures, from sampling to the final calculation and reporting of the results with the control of valid results. It is emphasized that the IQC program should be proportional and respond to the number of tests and the impact of the validity of the results, while the control program should respond to the number of analyses, according to clause 16.2 of the 2024 version. At the same time, various process controls are listed and described clearly, such as blanks, positive controls, negative controls. As for the selection of strains used as positive controls, these should be easily distinguishable so that they can be identified in the event of cross-contamination. For example, the use of rare or genetically modified strains is also recommended, e.g., fluorescent strains. For qualitative methods, positive controls should be used at a concentration close to the LOD₉₅ of the test method (e.g., 3 to 10 CFU/test portion) to avoid false-negative results, in accordance with clause 16.2.2.3 [2, 3, 13–15].

The application of IQC is not simply an analysis tool but a structured system that allows one to see whether the process was implemented correctly or whether it went out of control. In addition, IQC is a daily verification mechanism [2, 3, 17]. Critical factors that can affect the validity of the results are monitored, such as reagents, culture media, negative and positive controls, blanks, reference strains, and the interpretation of the results. In food chain samples to be microbiologically analyzed, IQCs are particularly important factors, since the results of microbiological analyses are affected by

pre-analysis procedures, by the heterogeneity of the microbial flora in the samples, by low target microbial populations, low concentrations of microorganisms, the presence of competing flora, and the risk of false-positive or false-negative results [1]. The implications of the revised standard for the use of positive and negative controls reinforce the view that quality control is no longer just an observation but a diagnostic tool for what went wrong. However, for small laboratories, this change also seems to be financially burdensome, as the frequency is based on a risk assessment. And while this standard provides flexibility in the frequency of IQC based on risk, for laboratories that do not have mature quality systems or interpret these systems incorrectly, the implementation of IQC can lead to gaps and inconsistencies for these laboratories [2, 13, 15, 17].

Technical changes and improvements of practical interest

In addition to the changes mentioned, the new standard also introduces new technical improvements that may be less apparent on first reading. Specifically, new alternative methods of confirmation and identification are recognized, a greater flexibility in the use of culture media under specific conditions is provided, and the possibility of using a larger inoculation volume in cases of low concentration of target microorganisms is permitted [2, 8, 9, 16]. The revised standard also recognizes the increasing use of automated systems for defined-volume handling, including automated dilution and pipetting devices and spiral plating equipment. Their use should be supported by appropriate performance verification, routine monitoring, and quality control procedures to ensure the accuracy and reproducibility of microbiological examinations.

Regarding the recognition of alternative methods of confirmation and identification beyond conventional biochemical and serological tests is that confirmation and identification can also be performed using appropriate techniques such as mass spectrometry and sequencing, provided that these are validated based on ISO16140 standard. This part is a significant update of the standard, as it incorporates the possibility of using new technologies and new equipment that many modern laboratories have [2, 9, 16]. A new improvement practice proposed by the standard is the use of ready-to-use (RTU) media. RTU media require reduced testing in the official laboratory, when they have been previously tested by the certified supplier based on ISO 11133. The concern here lies in the question of whether this is sufficient. This is sufficient provided that appropriate transport, storage, and handling conditions are maintained. In practice, during a laboratory audit, the laboratory should be able to demonstrate that RTU media and consumables have been transported, handled, and stored under the prescribed condition.

Table 2 Main differences between ISO 7218:2007 and ISO 7218:2024 and their implications for official control food microbiology laboratories

Aspect	ISO 7218:2007	ISO 7218:2024	Implications for official control laboratories	ISO 7218:2024 clauses
General approach	Descriptive guidance for microbiological examination	A more explicit risk-based and process-oriented approach	Strengthens control of factors affecting result validity across the entire examination process	1; 16.1
Transport and storage conditions	Temperature recommendations are more diverse and less consistent	More specific and consistent temperature categories for various sample types	Improves comparability of results and reduces ambiguity in pre-analytical handling Pre-analytical handling improves comparability of results and avoids uncertainty	9.2 to 9.4.2
Shelf-stable products	Transport: ambient temperature (< 40 °C) Storage before examination: 18 to 27 °C	Transport and storage before examination: room temperature (18 to 27 °C)	Narrows the recommended transport temperature range Introduces a more uniform permissible range for transport and storage	9.2; 9.4.2
Other products not stable at room temperature	Transport: 1 to 8 °C; storage before analysis: 3 °C ± 2 °C	Transport/storage: 5 °C ± 3 °C	Simplifies handling criteria, but demands more stringent temperature monitoring and documentation	9.2; 9.4.2
Surface samples	Reference to ISO 18593 and ISO 17604	Storage/transport between 1 °C and 8 °C, with reference to ISO 18593 and ISO 17604	Sets explicit requirements for swab and surface sample management	9.2 to 9.4.2
Freezing before testing	Less directly stated as a general restriction. Freezing was allowed as an exceptional measure when testing deadlines could not be met, provided that recovery of the target microorganism was shown not to be significantly impaired	Freezing before testing is not permitted, except in the case for previously frozen products	A substantive difference, not just a clarification Encourages the maintenance of microbiological integrity and does not permit the use of storage as a routine fallback	9.3
Retesting of stored samples	Less explicitly limited	Retesting of stored laboratory samples is not accepted practice; if performed retest findings must be reported separately as a retest	Strengthens scientific defensibility and reduces improper confirmatory testing	9.4.4
Laboratory organization and workflow separation	Good laboratory practice orientation	More precise standards for activity separation and contamination protection	Contamination control becomes more auditable and structurally integrated	4.4.2; 4.5.1 to 4.5.4; 10.1.2; 10.1.6
Internal quality control	More narrow or less clearly throughout the process	Expanded process-oriented IQC covering the full examination pathway	Positions IQC as an essential part of result validity rather than an extra task	16.1 to 16.2
IQC proportionality	Less clearly linked to risk and testing frequency	IQC should be proportionate to the impact and frequency of testing	Requires laboratories to justify control intensity using risk-based reasoning	16.2.1

Table 2 (continued)

Aspect	ISO 7218:2007	ISO 7218:2024	Implications for official control laboratories	ISO 7218:2024 clauses
Control materials	Standard use of controls	Increased emphasis on blank samples, positive and negative controls, and process control organisms	Enhances the ability to detect loss of control and identify process weaknesses	10.1.6; 14.1 to 14.3; 16.2.2
Confirmation and identification of methods	Mostly standard biochemical and serological methods	Increased acceptance of validated/verified new technologies like mass spectrometry and DNA sequencing	Promotes laboratory upgrading while keeping verification requirements	13.1 to 13.4; 17.3 to 17.4
Ready-to-use culture media	Increased reliance on testing by the user laboratory	When supplier testing and storage/transport control are guaranteed, less user testing might be appropriate	Focuses more on incoming quality assurance and supplier qualification	8
Low-count samples	Less explicit flexibility	Permits higher inoculum volumes while maintaining the inoculum-to-medium ratio and alternative methods for low microbial concentrations	Enhances the management of samples with low expected levels of contamination	10.2.2; 11.1; 11.2.2; 11.2.3.2
Quantitative calculations	Older calculation framework	Simplified calculations with calculator-based tools and clearer counting rules	Improves consistency, transparency, and reproducibility of reported results	11.2.6 to 11.2.7; 11.3.7 to 11.3.9; Annex B
Reporting and traceability	Less explicit contextual reporting	Requires clearer documentation of deviations, incidents, and uncertainty and the year of publication of the applied standard	Produces more transparent and decision-ready reports for regulatory use	9.1.2; 9.3; 10.1.5; 15

In this way, reduced controls by the laboratory for RTU materials are allowed. The revision therefore permits the reduced control of RTUs, transferring responsibility from the supplier to the laboratory, as long as the laboratory can demonstrate that the appropriate conditions were adhered to from transport to use [2, 8].

The revised standard is also more flexible in relation to the examination of samples expected to contain very low microbial numbers. For low-count situations, ISO 7218:2024 allows an increase in inoculum volume, including volumes greater than 1 mL in pour plating provided that the inoculum-to-medium ratio is maintained, and also recognizes approaches such as concentration or immunoseparation where relevant. In the spread-plate context, the limit of detection may be improved by testing 1.0 mL of sample or initial suspension either on a 140 mm plate or across three 90 mm plates. This is a useful clarification because it acknowledges the analytical difficulty of enumerating very low microbial populations while preserving methodological structure [2, 10, 15].

Quantitative calculations and reporting: simplification with tighter interpretive discipline

The simplification of the calculations section and the addition of calculator-based tools may seem like a technical convenience, but their significance is broader. Compared with ISO 7218:2007, the 2024 edition provides revised guidance for spiral plate counting and a simplified calculation framework, supported by clearer counting rules and calculator-based tools. These changes aim to reduce variation in the selection of countable plates, calculation, rounding, and reporting of quantitative results. Quantitative microbiology is one of the areas in which inconsistency in calculation, rounding, acceptance of plate counts, and interpretation of special cases can lead to substantial variation between analysts and laboratories.

The improved technique promotes more uniform handling of valid counts, low count circumstances, duplicate plates, serial dilutions, and confirmatory enumeration, improving not only numerical consistency but also clarity in how the final result was obtained [2, 14, 15]. The significance of the redesigned framework extends beyond the simplification of mathematics by decreasing the space for informal or ad hoc interpretation. Numerical defensibility is important in official control when using microbiological results to determine conformity. In addition to being technically valid, a result must be explainable, reproducible, and traceable in accordance with established calculation criteria. By connecting calculation practice to explicit acceptability criteria and reporting requirements, ISO 7218:2024 makes quantitative microbiology less dependent on tacit expertise alone and

more firmly embedded in verifiable laboratory rules [2, 14, 15].

Reporting and traceability: making the laboratory record more decision-ready

Finally, according to the standard, analytical findings should be linked to the conditions under which they were generated. In particular, it is required to include the year of publication of the applicable standard, the deviations as well as the uncertainty estimates, where necessary. This requirement is not only administrative but also demonstrates that the microbiological result has been obtained within a transparent methodological framework [2, 12].

In the test report, the presence of the year of publication of the standard enhances the transparency, clarity, and traceability of the methodological framework. Finally, the unsatisfactory condition of the sample or the insufficient sample or the re-examination of the sample helps to avoid incorrect interpretation of the results, while improving not only the laboratory report but also the quality of decision-making [2, 4, 12].

Discussion

In this manuscript, the authors do not suggest that the technical implementation of ISO 7218:2024 is fundamentally different in official and non-official laboratories. Many of the same requirements and operational challenges apply to both, particularly where laboratories are accredited according to ISO/IEC 17025. However, official control laboratories operate within a specific statutory framework and under formal designation by the competent authority, which may increase the regulatory significance of laboratory findings and the need for harmonized and auditable implementation.

The process-oriented approach of ISO 7218:2024 is consistent with the wider laboratory quality management system (LQMS) framework described by Pillai and Fox [19], who highlight that reliable laboratory outcomes depend on the management of the entire workflow, including pre-analytical, analytical, and post-analytical activities, rather than on the analytical method alone. In this context, the 12 core elements of the quality system—like organization, facilities and security, personnel, equipment, supplies and inventory, process management, documents and records, information management, evaluations, incident management, customer satisfaction, and continuous improvement—provide a practical structure for integrating quality into routine laboratory operations. In this respect, the revised ISO 7218:2024 can be interpreted as an expression of a broader LQMS logic, specific to microbiology, because it translates the general quality principles into documented controls for specimen

handling, contamination prevention, equipment and media management, internal quality control, reporting, and continuous improvement [19].

This is crucial for official control laboratories, where the result has both scientific and legislative weight. In this context, the revised standard supports three main objectives: First, it enhances harmonization between laboratories, reducing ambiguity on points such as transport temperatures, calculation practices, and decision rules. Second, it strengthens scientific advocacy, clarifying limitations on practices such as freezing before testing or retesting stored samples, a practice that many follow “for good reason.” Third and very importantly, it promotes a more mature quality culture: It promotes contamination control, internal quality control, and analytical deviation as an integral part of validity and not as optional good practices [2–4, 12, 14].

Thus, the success of the implementation of the new ISO 7218:2024 depends not so much on the narrow wording of its scope, but more on understanding the feasibility of its implementation. Measures such as improved workflow separation, appropriate IQC programs, effective sample handling schedules, and contamination prevention require sound organizational planning and management. Measures such as improved workflow separation, appropriate IQC programs, effective sample handling schedules, and contamination prevention require sound organizational planning and management. Laboratories that have a mature quality system, proper infrastructure, and sufficient staff can implement these modifications quite easily and smoothly. However, for smaller official control food laboratories, its correct implementation may be more difficult. In such cases, possible misinterpretations should be prevented by a good and strong technical supervision supporting the laboratory, as well as good accreditation control [2, 3]. The principal technical, quality assurance, and regulatory considerations associated with the application of ISO 7218:2024 in official control food microbiology laboratories are summarized in Table 3. Another item to consider is how ISO 7218:2024 restructures responsibility within the laboratory system. Instead of depending exclusively on post hoc correction or expert judgment to ensure result validity, the revised standard requires more preventive control at an earlier stage. This is illustrated by provisions concerning sample receipt, the restriction on freezing before testing, the limitations on retesting stored samples, and the need for root-cause analysis when IQC indicates that the testing process is out of control or when problems related to staff performance are identified. This strategy is scientifically robust since it resolves validity risks as they arise. However, it also implies that the laboratories must become more proactive in work planning, prioritization, and risk escalation prior to analytic procedures [2–4]. The changed standard has repercussions outside the laboratory. Since official control testing is integrated into larger

administrative and regulatory procedures, the scientific principles outlined in ISO 7218:2024 may challenge assumptions held by clients, inspectors, or enforcement organizations. The explanation that a preserved sample should not be routinely tested once more, for instance, is scientifically accurate, but it could conflict with expectations of confirmatory repetition. Similarly, tougher pre-analytical controls may necessitate closer collaboration with sample authorities and transportation providers. Thus, implementing ISO 7218:2024 should not be viewed just as the responsibility of the laboratory; it is also necessary to match the entire official control chain with microbiological principles [2, 4].

The practical implications for official control laboratories are threefold: i) more harmonized pre-analytical and computational practices, ii) clearer restrictions on freezing and retesting, and iii) stronger integration of contamination control and IQC into validity assurance. Comprehensive implementation will depend on infrastructure and equipment, personnel, technical supervision, and coordination throughout the official control chain. The main considerations are summarized in Tables 2 and 3 [2–4, 12, 14].

Furthermore, the new standard allows for the use of automated digital colony counters, recognizing their capacity to increase throughput and reduce inter-operator variability, hence enhancing overall data precision in microbiological testing [2, 17].

Section “[Technical changes and improvements in practical interest](#)” describes larger inoculation volumes and alternative concentration approaches for samples with low expected microbial populations, while Sect. “[Quantitative calculations and reporting: simplification with tighter interpretive discipline](#)” provides the corresponding calculation and reporting framework. Overall, these provisions support the selection of the method appropriate for the purpose, transparent calculation, and careful interpretation of low colony counts near the limits of detection or quantification [2, 17, 20–23]. The 2024 edition explicitly states that when low microbial counts are expected, the volume of inoculum in the pour plate may be increased, provided that the inoculum-to-medium ratio is maintained. The 1.0-mL spread-plate configurations were already present in ISO 7218:2007/Amendment 1:2013. Thus, the essential change of 2024 is the new explicit flexibility of the pouring slab together with the revised quantitative calculation framework described in Sect. “[Quantitative calculations and reporting: simplification with tighter interpretive discipline](#)” [2, 6, 7].

Conclusions

ISO 7218:2024 significantly strengthens the basic framework guiding microbiological tests in the food chain. Its significance stems from the adjustment of specific technical

Table 3 Legislative and accreditation requirements relevant to ISO 7218:2024 considerations for official control food microbiology laboratories

Category of requirements	Criterion	Provision/References	Technical & Legal Implication
Legislative	Accreditation	Regulation (EU) 2017/625, Art. 37(1), (4)-(5); ISO 7218:2024, Clause 1	To establish technical competence, a national accrediting organization (e.g., ESYD) must provide mandatory accreditation under ISO/IEC 17025. <i>“Official laboratories are designated by the competent authority, provided by the Regulation, operate and are accredited in accordance with EN ISO/IEC 17025 for the methods required.”</i> ISO 7218:2024 complements the accreditation framework by providing specific technical guidance for microbiology related to compliance with ISO/IEC 17025
Accreditation	Personnel Competence	ISO/IEC 17025:2017, Clause 6.2; ISO 7218:2024 Clause 16, Clause 5	Continuing training, technical proficiency evaluations, and authorized staff for specialist microbiological analysis. ISO 17025:2017 specifies general competence requirements, while ISO 7218:2024 provides specific provisions on personnel competence and verification of competence
Legislative	Proficiency Testing (PT)	Regulation (EU) 2017/625, Art. 38(2) ISO 7218:2024 Clause 16.3	Participation in interlaboratory comparisons is required to ensure the accuracy and reproducibility of results against peer labs. <i>“Upon request by the relevant EU reference laboratory or national reference laboratory, official laboratories participate in interlaboratory comparative tests or proficiency tests for the analyses, tests or diagnoses they perform as official laboratories.”</i> In ISO 7218:2024 PT schemes assess laboratory performance through PT or interlaboratory comparisons; individual performance results should be evaluated as well as trends
Legislative	Methods used for official controls	Regulation (EU) 2017/625, Art. 34	<i>“Methods used for official controls shall comply with Union rules establishing those methods or the performance criteria for those methods.”</i> In their absence, official laboratories use methods according to Article 34(2)-(3)
Accreditation	Calibration & Equipment	ISO 7218:2024, Clause 6; ISO/IEC 17025:2017, Clauses 6.4 to 6.5	Strict metrological traceability, including regular calibration and performance verification of equipment (e.g., incubators, autoclaves)
Legislative	Method Validation and/or Verification	Regulation (EU) 2017/625, Art. 34; ISO 7218:2024, Clause 17	Addresses validation and verification of microbiological methods and refer to the ISO 16140 series for detailed procedures; Ensures that methods are “fit-for-purpose” by providing documented evidence of sensitivity, specificity, and limits of detection (LOD)
Accreditation	Measurement of Uncertainty (MU)	ISO/IEC 17025 Clause 7.6; ISO 19036:2019; ISO 7218:2024 Clause 11.4, Clause 15, Annex B	Quantifying statistical doubt in the outcome is critical for assessing compliance with microbiological standards (Reg. 2073/2005). Measurement uncertainty is evaluated as applicable to the testing activity; ISO 19036 provides the framework for quantitative food microbiology

regulations, as well as from the broader effort to manage laboratory conditions that are most likely to affect result validity. In this regard, the new edition prioritizes pre-analytical integrity, contamination prevention, internal quality control, reporting, and risk-based laboratory management.

For official control microbiological laboratories, the updated standard offers clear advantages. It promotes practice consistency, facilitates improved comparison between laboratories, and strengthens the scientific and regulatory defensibility of microbiological findings. Particularly important are the revised requirements regarding sample transport and storage, the provision that freezing before examination is not permitted, the clarified limitations of retesting stored samples, the more structured treatment of workflow segregation, the broader acceptance of alternative confirmation technologies, and the expanded concept of IQC. However, it is unlikely that implementation will be consistent or simple. The updated standard enhances organizational capacity as well as analytical proficiency. Adequate infrastructure, well-thought-out protocols, skilled personnel, thorough documentation, and the capacity of laboratories to conduct meaningful risk assessments related to their own operations will all be necessary for its successful implementation. Because of this, the adoption of ISO 7218:2024 should be viewed as a managed shift in laboratory practice rather than a straightforward document update.

To sum up, ISO 7218:2024 should be considered a crucial tool for enhancing the uniformity, dependability, and transparency of microbiological testing in official control settings. The degree to which laboratories can translate their principles into consistent, auditable, and scientifically sound routine practice will determine their true worth rather than just formal compliance.

In conclusion, this article offers a targeted and timely assessment of the application of ISO 7218:2024 in microbiological laboratories for official food control, highlighting the importance of the new edition not only as a technical standard, but also as a tool for enhancing laboratory quality, reliability of results, and harmonization of official controls. Its main contribution lies in linking the new edition of the standard to the practical needs of accredited laboratories and producing results with direct relevance for consumers' protection and public health and compliance with legislation. At the same time, by comparing it with the previous edition, ISO 7218:2007, the article contributes to a better understanding of the changes introduced by ISO 7218:2024 and their potential impacts on sample management, the application of microbiological methods, documentation, traceability, and the overall operation of laboratory quality systems. However, this work has certain limitations, as it is mainly based on a review of the literature and regulatory documents rather than on primary empirical data. This work is a Perspective Review based on a critical analysis of

standards, legislation, and relevant literature rather than on primary data collected from laboratory personnel, accreditation assessors, competent authorities, or other stakeholders. No interviews were conducted with laboratory personnel and competent authorities, nor was a quantitative approach through questionnaires applied to capture the readiness, needs, or difficulties of implementing ISO 7218:2024 in different laboratory environments. Therefore, the conclusions of the work should be interpreted as a documented analytical assessment of the standard that requires further evaluation under real-world implementation conditions. Future steps could include field studies in official testing laboratories, comparative evaluation of the implementation of the standard between different countries or laboratory systems, development of self-assessment tools for the implementation of ISO 7218:2024, and investigation of the educational, technical, and organizational needs arising during the transition from the previous to the new version of the standard.

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Data availability No datasets were generated or analyzed during the current study.

Declarations

Conflict of interest The authors declare that they have no known competing financial or personal interests that could have appeared to influence the work reported in this paper.

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References

1. World Health Organization (WHO) (2011) Laboratory quality management system: handbook. World Health Organization, Geneva
2. International Organization for Standardization (ISO) (2024) ISO 7218:2024. Microbiology of the food chain—General requirements and guidance for microbiological examinations. Geneva, Switzerland: ISO
3. International Organization for Standardization/International Electrotechnical Commission (ISO/IEC) (2017) ISO/IEC 17025:2017. General requirements for the competence of testing and calibration laboratories. Geneva, Switzerland: ISO
4. European Parliament and Council of the European Union (2017) Regulation (EU) 2017/625 of 15 March 2017 on official controls and other official activities performed to ensure the application of food and feed law, rules on animal health and welfare, plant health and plant protection products. Off J Eur Union L95:1–142
5. World Health Organization (WHO) (2020) Laboratory biosafety manual, 4th ed. WHO, Geneva
6. International Organization for Standardization (ISO) (2007) ISO 7218:2007. Microbiology of food and animal feeding stuffs—General requirements and guidance for microbiological examinations. Geneva, Switzerland: ISO
7. International Organization for Standardization (ISO) (2013) ISO 7218:2007/Amd 1:2013. Microbiology of food and animal feeding stuffs—General requirements and guidance for microbiological examinations—Amendment 1. Geneva, Switzerland: ISO
8. International Organization for Standardization (ISO) (2014) ISO 11133:2014, including Amd 1:2018 and Amd 2:2020. Microbiology of food, animal feed and water—Preparation, production, storage and performance testing of culture media. Geneva, Switzerland: ISO
9. International Organization for Standardization (ISO) (2005) ISO 22174:2005. Microbiology of food and animal feeding stuffs—Polymerase chain reaction (PCR) for the detection of food-borne pathogens—General requirements and definitions. Geneva, Switzerland: ISO
10. International Organization for Standardization (ISO) (2017) ISO 6887–1:2017. Microbiology of the food chain—Preparation of test samples, initial suspension and decimal dilutions for microbiological examination. Geneva, Switzerland: ISO
11. International Organization for Standardization (ISO) (2018) ISO 18593:2018. Microbiology of the food chain—Horizontal methods for surface sampling. Geneva, Switzerland: ISO
12. International Organization for Standardization (ISO) (2015) ISO 17604: 2015. Microbiology of the food chain—Carcass sampling for microbiological analysis. Geneva, Switzerland: ISO
13. International Organization for Standardization (ISO) (2019) ISO 19036:2019. Microbiology of the food chain—Estimation of measurement uncertainty for quantitative determinations. Geneva, Switzerland: ISO
14. International Organization for Standardization (ISO) (2010) ISO/TS 22117:2010. Microbiology of the food chain—Specific requirements and guidance for proficiency testing by interlaboratory comparison. Geneva, Switzerland: ISO
15. International Organization for Standardization (ISO) (2005) ISO 14461–2:2005. Milk and milk products—Quality control in microbiology laboratories—Part 2: Determination of the reliability of colony counts and proficiency testing in colony count procedures. Geneva, Switzerland: ISO
16. International Organization for Standardization (ISO) (2016) ISO 16140–2:2016. Microbiology of the food chain—Method validation. Part 2: Protocol for the validation of alternative (proprietary) methods against a reference method. Geneva, Switzerland: ISO
17. Magnusson B, Tsimillis KC, editors (2023) Accreditation for microbiological laboratories. Eurachem Guide. 3rd ed. Eurachem. ISBN 978–91–519–6581–9. Available online at: https://www.eurachem.org/images/stories/Guides/pdf/Eurachem_Guide_AML_2023_EN.pdf.
18. World Health Organization (WHO) and Food and Agriculture Organization of the United Nations (FAO) (2016) Statistical aspects of microbiological criteria related to foods: a risk manager's guide. Microbiological Risk Assessment Series No. 24. Geneva: WHO
19. Pillai SP, Fox E (2025) Laboratory quality management system fundamentals. Front Bioeng Biotechnol 13:1578654. <https://doi.org/10.3389/fbioe.2025.1578654>
20. Parratt K, Newton D, Dunkers J, Dootz J, Hunter M, Logan-Jackson A, Pierce L, Sarkar S, Servetas SL, Lin NJ (2025) Measurement quality metrics to improve absolute microbial cell counting. Front Microbiol 16:1631377. <https://doi.org/10.3389/fmicb.2025.1631377>
21. Fujikawa H, Tsubaki H (2019) Characteristics of microbial colony counts on agar plates for food and microbial culture samples. Food Hyg Saf Sci (Shokuhin Eiseigaku Zasshi) 60(4):88–95
22. Morton RD (2001) Aerobic Plate Count. In: Compendium of Methods for the Microbiological Examination of Foods. APHA Press
23. Chik AHS, Schmidt PJ, Emelko MB (2018) Learning something from nothing: the critical importance of rethinking microbial non-detects. Front Microbiol 9:2304. <https://doi.org/10.3389/fmicb.2018.02304>

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